

Constraint Collapse and Fidelity Decay

When Feedback Stops Correcting Symbolic Systems

SFP 04 | Semantic Fidelity Project

A. Jacobs

Abstract

Symbolic systems can continue receiving feedback after feedback has lost the power to enforce correction. This paper defines constraint collapse as a condition in which feedback, consequence, and correction remain present but no longer bind representations or behavior to external conditions. It explains how semantic fidelity decays when detection is separated from revision and why continued operation can conceal deepening misalignment.

Core Claim

A system enters constraint collapse when detected error no longer has the authority to interrupt continuation or alter the representation that produced it.

Definition

Constraint collapse is a systemic condition in which feedback, consequence, and correction remain present in symbolic systems but lose their ability to enforce change. The system continues to operate and update representations, yet no longer possesses reliable mechanisms for invalidating error or stopping misaligned processes.

This differs from the absence of feedback. Complaints may be logged, contradictions documented, audits completed, and corrections issued. The failure lies in their binding force. Information about error circulates without changing the structure that continues to produce it.

Constraint and Correction

A constraint links representation to consequence. It gives external conditions the power to reject, revise, or halt an internal model. Effective constraint does more than supply information. It changes what the system can continue doing.

When constraint weakens, correction becomes advisory. The system can acknowledge a discrepancy while preserving its operating assumptions. It can satisfy a review procedure without revising the model under review. Responsiveness remains visible, but contact no longer governs continuation.

How Collapse Develops

Constraint collapse commonly develops through delayed consequence, externalized cost, procedural substitution, and representational distance. The system receives signals after the relevant decision, transfers the cost to actors outside the feedback loop, treats completion of a corrective procedure as correction itself, or evaluates representations against other internal representations.

Compression intensifies the problem. As the referent becomes more distant, the system relies on dashboards, summaries, classifications, and model outputs to determine whether it remains aligned. These representations are easier to update than the conditions they describe. Correction gradually shifts from changing reality-facing behavior to changing the account of that behavior.

The Sequence of Noncorrection

The system may detect an error, register the feedback, and complete whatever corrective process is required without changing the representation that caused the problem. On paper, the system has responded. In practice, nothing important has been revised.

That is where correction can become procedural rather than corrective. Reports are filed, reviews are completed, and responses are documented, but the underlying model remains intact. Over time, the system can become very good at handling evidence of error without allowing that evidence to change what it does.

Constraint Collapse and Fidelity Decay

Semantic fidelity depends on continued access to a referent and on consequences that make divergence costly. Once feedback loses binding force, representations can remain fluent and internally consistent while their connection to the referent weakens.

Fidelity decay is cumulative. Early losses may affect nuance or context. Later losses alter causality, category boundaries, and the criteria by which the system evaluates itself. Because each new representation is built from earlier ones, the system can become more coherent as its external alignment declines.

The Drift Principle

The Drift Principle states that representations can be compressed, optimized, distributed, and reused faster than reality can correct them. Constraint collapse describes the condition that allows this imbalance to persist.

When optimization accelerates and corrective contact weakens, a system improves its representations according to internal criteria. Performance indicators may rise because the system has become more effective at satisfying the measures through which it sees itself. Continued functionality then conceals the loss of external correction.

Distinguishing Adjacent Failures

Feedback absence means the system receives no corrective signal. Feedback weakening means the signal arrives with reduced speed, clarity, or authority. Verification substitution means the system checks whether a procedure was followed instead of whether the underlying condition is true. Constraint collapse is the broader state in which available correction can no longer force revision or halt continuation.

This distinction matters because adding more feedback does not repair a system whose feedback lacks authority. More reports, reviews, and metrics may increase the appearance of correction while strengthening the representational layer that insulates the system from consequence.

Design Requirements for Correction

Systems that preserve constraint require stopping conditions, access to original evidence, consequences attached to persistent divergence, and correction that propagates into future behavior. Human review functions as a constraint only when the reviewer can change or stop the process and when that intervention alters what happens next. A correction mechanism should therefore be judged by its effect, not by its presence. The relevant test is whether the system behaves differently after the discrepancy becomes known.

Conclusion

Constraint collapse explains how a system can remain responsive without remaining correctable. Feedback continues, procedures continue, and outputs continue. The missing element is the power of reality to invalidate the representation. The absence of failure is not proof of alignment.